

Integrated Experiments – Solution

Experiment 1: Decision Tree Classification

The Iris dataset is loaded from scikit-learn. It contains four numerical features and three classes. Missing values are checked; the original dataset contains none. The data is divided into 80% training and 20% testing data using stratification.

A DecisionTreeClassifier with criterion='gini' is trained. The first feature used by the fitted tree is the root node. It is selected because it gives the greatest impurity reduction at the root under the Gini criterion.

The model is evaluated using accuracy, confusion matrix and classification report. The executable code prints the actual misclassified test samples, so no fabricated errors are used.

Experiment 2: Q-Learning

A 4×4 Grid World uses state 0 as Start, state 15 as Goal, and states 5 and 10 as obstacles. Actions are Up, Down, Left and Right. Rewards are +10 for the goal, −1 for a normal step and −5 for an obstacle.

Hyperparameters are $\alpha=0.1$, $\gamma=0.9$, initial $\epsilon=1.0$, minimum $\epsilon=0.05$, decay=0.97, and 100 episodes. The standard Q-learning update is applied after every action.

Q-values for states 0 and 6 are recorded at Episodes 1, 50 and 100. The final policy chooses the action with the maximum Q-value at each non-terminal state.